

Answers

VARC

1.D	2.2	3.D	4.D	5.D	6.B	7.B	8.B
9.B	10.A	11.A	12.4	13.D	14.D	15.D	16.B
17.B	18.A	19.B	20.B	21.D	22.D	23.A	24.D

LRDI

25.4	26.A	27.C	28.2	29.26	30.B	31.C	32.5100
33.3500	34.150	35.A	36.4	37.C	38.C	39.D	40.D
41.B	42.B	43.4	44.5	45.6	46.B		

Quant

47.2160	48.2	49.340	50.14	51.A	52.B	53.C	54.A
55.A	56.A	57.139	58.A	59.D	60.A	61.D	62.A
63.7	64.11	65.7	66.B	67.B	68.B		

Explanations

VARC

1. D

D) Option 2 is the correct answer.

The sentence before option 2 states "Now, however, we are having the last laugh". The reason they are having the last laugh is given in the sentence, "*Science has officially crowned us superior to our early-rising brethren.*" Therefore, it fits here perfectly forming a coherent paragraph.

A) Option 4: Placing the sentence here would seem a bit disconnected. The sentence before and after Blank 4 flows continuously and placing the sentence here would break this flow

B) Option 3: This placement would disrupt the flow of the sentence before and after Blank 3. The study mentioned before, blank 3, is the moment where the night owl's position is validated, so introducing the idea of "superiority" will be out of place and feel redundant.

C) Option 1: It may seem plausible initially, but it would not be right to talk about the claim without first mentioning the negative stereotypes, making this placement less effective.

[VIDEO SOLUTION](#)

2.2

Sentence 2 is the odd one out.

Sequence 1435 forms a coherent paragraph:

Sentence 1 introduces the debate about gender being influenced by both biological and social factors. It sets the tone for the paragraph, addressing the complexity of gender.

Sentence 4 follows logically from Sentence 1. After introducing the complexity of gender, it suggests that, even with the acknowledgment of both nature and nurture, researchers studying gender's biological basis still face political challenges.

Sentence 3 builds on Sentence 4, highlighting why researchers studying the biological aspects of gender face resistance. The public discomfort with viewing gender as more than just a social construct is explained.

Sentence 5 connects to Sentence 3 by elaborating on the political preference for viewing gender purely as a social construct, reinforcing the challenges faced by researchers advocating for a more nuanced view of gender.

Sentence 2 doesn't fit in the context of the paragraph. The rest of the sentences are focused on the complexities of gender as a mix of nature and nurture and the political and societal challenges researchers face. Sentence 2 shifts the focus to the issue of sexism, which, while related to gender, does not directly tie into the previous sentences discussing the nature vs. nurture debate and the political resistance to biological explanations.

VIDEO SOLUTION

3.D

We can attempt to place the given sentence by breaking the passage down into three segments -

The discussion till Blank (2) can be considered the first segment: it introduces the idea that globalization was already underway well before the industrial era. It highlights the economic interconnectedness between Chinese consumers and Southeast Asian suppliers, emphasizing that trade networks and economic reorientations were already shaping societies. The missing sentence does not fit here in Blanks (1) or (2) because the focus is on early economic exchange rather than the role of Europeans in globalization.

The discussion between Blanks (2) and (3) can be labelled as the second segment: it explores how trade routes developed in the year 1000, predating Columbus's journey. It sets up a comparison between different historical periods - 1000 and 1492 - but the idea this comparison is leading to is still unclear.

This is where the final segment [the discussion after Blank (3)] fits in. It delivers the passage's key argument: European involvement in global trade was not a beginning but a continuation of pre-existing systems. It reinforces the idea that the economic structures and trade routes established before 1492 were instrumental in enabling European expansion. The given sentence would best fit Blank (3) because we make a declaration ["The Europeans did not invent globalization"] and then follow up with information that reinforces this idea ["They changed and augmented what was already there since 1000"].

Hence, Option D is the correct choice.

VIDEO SOLUTION

4.D

The passage starts by explaining how states use the DIME model in warfare, often targeting information systems first to destabilize enemies. Governments monitor communication networks, raising privacy concerns. With social media's rise, many countries employ "cyber troops" to manipulate public opinion online, spreading disinformation through tactics like paid ads. Option D best captures all these points.

Option A: This option incorrectly focuses on open data policies being used to deploy cyber troops, which the passage does not emphasize.

Option B: This option fails to address the key part of the passage, which is monitoring telecom data and networks.

Option C: The passage does not indicate that governments conduct warfare simultaneously with military forces; rather, they act before involving the military. We can infer this from the statement, "Most states do everything they can to inflict pain and confusion on their enemies before deploying the military."

VIDEO SOLUTION

5.D

Option D best captures the essence of the passage.

John Cleese argues that comedians need freedom to be creative and that fear of offending people or worrying about the consequences of what they say hinders their ability to be original. He highlights how modern comedians often second-guess themselves, which stifles their creativity. Option D reflects this idea, which stresses the importance of freedom and creativity in comedy, while warning against the fear that stifles innovation.

Option A: While Cleese advocates for freedom in comedy, he doesn't say that comedians "must go where no one has gone before," which is more of an extreme interpretation than the essence of the passage.

Option B: This option focuses too much on the relationship between creativity and critical thinking. The passage is more about how fear of offending hinders creativity, not about creativity being incompatible with critical thinking.

Option C: This focuses on the "death of the art form," but Cleese's main point is about how fear of repercussions impacts creativity, not about the art form dying as such.

VIDEO SOLUTION



6.B

Evaluation of the options:

Option 1: The sentence starting with "Yet each day..." would not fit here. This part is introducing Kervran's initial observations about the hens. Inserting this sentence would feel disjointed because the paragraph hasn't yet explained the significance of the eggs.

Option 2: The sentence wouldn't fit here either. The focus here is on what the hens are doing (pecking mica), and the sentence about eggs feels disconnected from this point in the narrative.

Option 3: The sentence about the eggs fits perfectly here because it explains a key element of the puzzle: the hens are producing eggs with calcareous shells despite not having an obvious source of calcium. This provides a crucial piece of information that explains why Kervran was puzzled in the first place. It ties directly to the anomaly he is investigating.

Option 4: It won't fit here as the paragraph has already shifted to Kervran's conclusion about the chickens transmuting one element to into another. Inserting the sentence about the eggs here would introduce new information too late, disrupting the focus on Kervran's final findings.

Therefore, The sentence fits best in Option 3. It helps to explain Kervran's puzzling observations and contributes to the mystery that he's trying to solve, all while maintaining a smooth narrative flow. The other options either disrupt the flow or introduce new information too soon or too late.

VIDEO SOLUTION

7. B

Option B is the correct answer.

This is not an externality because it was the original intended use of air-conditioning. The passage mentions that Carrier invented air-conditioning to ensure cool, dry air for factories with low-humidity requirements.

The other options are externalities:

Option A: This is an unintended consequence. CFCs were initially used in refrigeration and air-conditioning, but their long-term environmental impact (ozone depletion) was not anticipated.

Option C: The Jacquard loom was originally a mechanical device for weaving patterns in fabric, but it led to the development of programmable machines, which had far-reaching effects on modern computing, which was also an unintended consequence of the loom's invention.

Option D: The passage states that the phonograph was initially designed for dictation but was adapted for music recording, which was an unintended consequence.

VIDEO SOLUTION

8. B

Option B is the correct answer.

The first paragraph discusses how technological or industrial advances are often accompanied by unintended consequences or secondary effects, which may not be fully understood or predictable at the time of the invention. It gives the example of the printing press, which led to unexpected developments such as the creation of spectacles, and later, the telescope and microscope. This is well captured in Option B.

Option A: While some secondary effects may be beneficial, the focus of the paragraph is more on the unpredictability and far-reaching nature of these effects, rather than their inherent benefit.

Option C: The paragraph doesn't advocate judging inventions by their secondary effects. Instead, it mentions that these effects are unpredictable and sometimes surprising. The main point is their unpredictability, not how to judge an invention.

Option D: The paragraph does not suggest that the impact of a technological advance should be evaluated by the boost its secondary effects give to generating further technological advances.

VIDEO SOLUTION

9. B

Option B is the correct answer.

The author mentions Carrier, Babbage, and Edison to emphasize that the inventors' original intentions were not related to the unexpected societal impacts their inventions had:

- Carrier created air-conditioning for industrial use, but it triggered a mass migration to cities like Phoenix and Las Vegas.
- Babbage's invention of a programmable loom and Edison's phonograph were originally intended for specific purposes (textile weaving and dictation, respectively). Still, they led to far-reaching technological developments in computing and music industries.

Therefore, we can infer that the inventors did not anticipate the full consequences of their inventions.

Option A: The secondary effects are shown as surprising or leading to unforeseen societal changes, rather than leading to more inventions.

Option C: The passage doesn't claim that the unintended consequences were largely beneficial. While some consequences may have been beneficial (like the telescope and microscope from the printing press), others (like climate change) have been harmful.

Option D: This is close, but it's not the main point. The intended purpose of the inventions may have been different from their actual use, but the author's primary argument is about how the inventors could not have predicted the full societal impacts of their inventions, rather than focusing on how inventions end up being used for different purposes.

VIDEO SOLUTION

10. A

Option A is the correct answer.

The author does not imply that recent advances are more threatening than past ones. Instead, he suggests that the nature of technological progress (with more individuals and smaller groups able to innovate) has changed, leading to new risks. The focus is not on comparing "past vs. recent" as more threatening but on the unexpected global impacts of all technological advances. Thus, option A misrepresents the author's view.

Option B: The author would agree with this because the passage explains that Ethyl (leaded fuel) and Freon are examples of innovations whose unintended consequences stem from by-products they emitted (such as lead from fuel and chemicals from Freon), which had secondary effects on health and the environment.

Option C: The author would support this as he mentions how individuals or small groups can now create innovations that have global impacts, particularly in fields like biotechnology and chemistry, which was less true in the past.

Option D: The author suggests that the health threats of leaded fuel were visible earlier (in the 1920s), implying that they should have been addressed sooner. Hence, the author would support this

CAT Percentile Predictor



11. A

Option A is the correct answer.

Option A captures the core idea that their collaborations have led to valuable scientific discoveries despite differences in research methods between physicists and other scientists. The passage emphasizes how these contrasting approaches still lead to productive outcomes, demonstrating the benefits of cross-disciplinary work.

Option B: While large data sets and mathematical models are mentioned, this option incorrectly focuses on "large data sets and mathematical models" as the main contributor, which is not the main point of the passage.

Option C: This is partially true, but the passage does not emphasize the "desire to diversify" research or focus on social scientists. It is about the collaboration of different scientific fields, not specifically social science.

Option D: This is inaccurate because the passage does not state that physicists have "buried" their differences; rather, it says that their differences exist, but the collaboration is still valuable. This is an extreme interpretation.

12. 4

Sentence 4 is the odd one out.

Sentences 1, 2, 3, and 5 all focus on the current status and future prospects of cultivated meat in the UK, such as the leadership in developing cultivated meat, regulatory assessments, and the importance of government investment in research.

Sentence 4, however, refers to the previous UK government's efforts to fast-track approval of cultivated meat which is disconnected from the other sentences, as the focus shifts to a past action rather than the current developments or future needs in the industry.

13. D

Option D is the correct answer.

The passage provides several arguments in favour of making peer review data public, but it does not mention that making this data public would help in selecting appropriately qualified reviewers for academic writing.

Option A: The passage mentions that openly publishing peer review data could help journals address fraudulent peer reviews, such as authors using fake email addresses to review their own manuscripts. Therefore, this is a reason given in favour of making the data public.

Option B: The passage discusses a study showing that women and non-Westerners are underrepresented among peer reviewers. Publishing peer review data could help highlight these gender and race biases, which is one of the reasons for making the data public.

Option C: The passage suggests that sharing peer-review data could help journals tackle issues like fraud, inefficiency, and systemic bias in the publishing process, which is another reason given in favour of making the data public.

14. D

Option D is the correct answer.

This is not mentioned as a reason in the passage. The focus is on staying informed, contributing to the field, and publicizing their own work, not on expanding influence. Therefore, option D is the correct answer as it is the only reason not mentioned in the passage.

Option A: The passage mentions that some academics review work to "*keep abreast with developments in their field*," which aligns with the idea that reviewing helps them stay updated with cutting-edge ideas. This is a valid reason.

Option B: The passage also mentions that some scientists post their reviews online "*mainly to claim credit for their work*," which indicates that some view reviewing as an opportunity to publicize their contributions. This is a valid reason.

Option C: The passage states that some view reviewing as "*a duty to the discipline*," which is a form of service to the academic community. Therefore, option C is also a valid reason.

15. D

Option D is the correct answer.

The passage discusses the value of making peer-reviewed data public to help address various issues in academic publishing, such as bias, inefficiency, and fraud. The author advocates for a shift "*from opacity in all cases to opacity only when necessary*," implying support for greater transparency in peer review.

Option A: The passage mentions concerns about reviewer selection, such as gender and geographic imbalances, but it does not suggest that careful screening of content-familiar reviewers is a priority.

Option B: The passage mentioned about maintaining confidentiality, but it does not strongly argue in favour of preserving anonymity.

Option C: The author does not fully endorse the publication of actual reviews or reviewer identities in every situation. While the author advocates for sharing peer review data, they also acknowledge that, in some cases, the content of reviews and reviewers' identities may need to remain confidential.

16. **B**

Let's evaluate the options:

Option A: The passage mentions this concern: "referees may be less critical of manuscripts if their reports are published, especially if they are revealing their identities by signing them."

Option B: The passage does not mention that one of the reasons to oppose open peer review is to protect reviewers from unwarranted and unjustified criticism. Instead, the passage discusses the concern that reviewers may avoid giving critical feedback if their identities and reports are made public, not because they want to avoid unjust criticism. The key concern is the fear of justified criticism rather than avoiding unjust or unwarranted criticism.

Option C: The passage mentions that reviewers may be less critical in their reports if they fear their reviews will be published. This might prevent reviewers from being honest and offering critical assessments, which are crucial for the integrity of the publishing process.

Option D: Another concern raised in the passage is that reviewers might take longer to submit their reviews if they know they will be publicly scrutinized. This delay in the review process is seen as a disadvantage of open peer review.

Therefore, **option B is the correct answer** because the passage does not mention concerns about leaving reviewers unexposed to unwarranted criticism.

[VIDEO SOLUTION](#)

17. **B**

Option B is the correct answer.

The demand for spices was not necessarily tied to their availability. The passage states that medieval Europeans had limited geographical knowledge of where spices came from and were highly prized despite their relative scarcity. Spices were used for culinary and medicinal purposes, and the demand was driven by cultural and medical factors rather than availability.

Option A: The passage states that spices were used partly due to the unavailability of refrigeration, and some spices served as antibacterial agents. With the development of refrigeration techniques, the necessity for spices to preserve or enhance food properties would diminish, leading to a decline in their appeal.

Option C: The passage mentions how spices were used for medicinal properties in medieval times as part of humoral balancing. As medical science evolved, such practices would likely fall out of favour, reducing their significance.

Option D: The passage states that spices played a significant role in medieval cooking, balancing humoral properties in food. Changes in European tastes or culinary practices would have contributed to a decline in the demand for spices.

[VIDEO SOLUTION](#)

18. **A**

Option A is the correct answer.

This is a wrong conclusion because it confuses the physical heat from the harvesting process with the pepper's actual spiciness. The process of using fire for harvesting could not be related to the spiciness (being hot) of the pepper.

Option B: The story of snakes and burning trees implies that pepper was difficult and dangerous to harvest, which could explain why it was costly. This is a reasonable conclusion based on the story.

Option C: Given the danger described in the myth (snakes and burning trees), it's logical that people would conclude it's not advisable to go to India to harvest pepper themselves. This could be a conclusion from the story.

Option D: The story suggests that harvesting pepper is difficult and dangerous, which might lead people to think that the supply is limited. Based on the information provided in the myth, this is a reasonable conclusion.

[VIDEO SOLUTION](#)

19. **B**

Option B is the correct answer.

While spices were a major part of European trade with the East, the passage does claim that gold was a motivation for colonizing India. The main focus of the passage is on spices, not gold. Hence, the conclusion that India was colonized for both spices and gold cannot be definitively drawn from the passage.

Option A: The passage hints that the desire for spices played a significant role in driving European colonial expansion. Therefore, this conclusion can be reached from the passage.

Option C: The passage briefly mentions that spices never had the same enduring allure or commercial potential as gold, silver, tobacco, indigo, or sugar. From this, we can infer that tobacco was more marketable than spices at certain points in history.

Option D: In the passage, the desire for spices is described as one of the major factors that led to European colonialism. Therefore, this conclusion could be reached.

[VIDEO SOLUTION](#)

20. **B**

Option B is the correct answer.

If white peppercorns were brought to Europe, Europeans would likely doubt the myth of harvesting pepper by burning trees, as white peppercorns would not be burnt like the black peppercorns described in the story. This inconsistency would make them question the accuracy of the myth.

Option A: Medieval maps were mentioned as symbolic and inaccurate, making them impractical for navigation. Traders did not rely on these maps for precise geographical guidance.

Option C: This outcome is unlikely. The passage explains that spices were scarce and expensive; thus, bringing white pepper would not significantly affect its price, since it remained a rare and highly valued commodity.

Option D: Even if a trader brought white peppercorns, pepper would still be viewed as exotic by Europeans due to its rarity and the long journey it took to reach Europe. So, this outcome is also unlikely.

[VIDEO SOLUTION](#)



21. **D**

Option D is the correct answer.

The passage states that: *"As they become more reliant on human food they might find at campsites or in rubbish bins, they become less avoidant of humans. Losing that instinctive fear response puts them into more situations where they could get into an altercation with a human, which often results in that bear being put down by humans."*

This means that the factor contributing most to carnivores' habituation to human presence is the reduction in instinctive fear response and reliance on human food sources, which is evident in option D.

Option A: The passage does not mention their natural aggression. It focuses more on losing fear due to food reliance on human food.

Option B: The passage mentions climate change as a possible reason for increased conflict, but it doesn't suggest it directly contributes to loss of fear or habituation.

Option C: The passage does not mention about the predatory perception of humans as potential prey, hence eliminated.

[VIDEO SOLUTION](#)

22. **D**

Option D is the correct answer. The passage mentions that landscape heterogeneity (a mix of farmland, forests, and carnivore habitats) in rural areas of low-income countries creates more opportunities for human-carnivore encounters.

As Penteriani's research shows, such landscapes increase the statistical probability of these conflicts because the areas are interspersed with human and carnivore habitats. This aligns with option D.

Option A: This is inconsistent with the passage. The passage states, *"The likelihood of human-carnivore conflicts appears to be higher in areas of low-income countries dominated by vast rural landscapes and farmland"*. Therefore, it is not less prone, rather more prone as per the passage.

Option B: The passage does not claim that landscape heterogeneity inherently decreases the chances of human-carnivore conflict. Instead, it states that diversity increases the likelihood of encounters rather than reducing it.

Option C: The passage does not state that homogeneous landscapes cause high rates of conflict due to predictability. Instead, it suggests that landscape heterogeneity increases encounters.

[VIDEO SOLUTION](#)

23. **A**

This is an official CAT 2024 Question, and the marked answer is according to the official answer key. We disagree with this answer.

Let us look at falsifying these statements:

Option A: Climate change has had negligible effects on the frequency of carnivore-human interactions in affected regions.

False version 1: Climate change has had no effect on the frequency of carnivore-human interactions in affected regions. - Inconsistent with the passage

False version 2: Climate change has had a lot of effect on the frequency of carnivore-human interactions in affected regions. - Consistent

As one of the versions contradicts the passage, we can say that the statement if false is inconsistent.

Option B:

Predatory attacks by carnivores are a common occurrence and have steadily increased over the past few decades.

False Version: Predatory attacks by carnivores are a rare occurrence and have steadily increased over the past few decades.

This is consistent with the passage.

Option C: Carnivores lose their instinctive fear of humans, when consistently exposed to human food sources.

False Version: Carnivores do not lose their instinctive fear of humans, when consistently exposed to human food sources. -- irrelevant to the passage as the author speaks on reliance on human food and not exposure to human food "sources".

Option D: Human efforts to avoid risky behaviours around large carnivores have proven effective in reducing conflict incidents.

False version: Human efforts to avoid risky behaviours around large carnivores have not proven effective in reducing conflict incidents. -- beyond the scope of the passage.

[VIDEO SOLUTION](#)

24. **D**

Option D is the correct answer.

The passage mentions that 66% of coyote attacks involve a dog, which can either provoke a carnivore or escalate a dangerous situation when the dog chases a carnivore or vice versa. In areas with large carnivores, unleashing dogs can increase the likelihood of encounters and conflicts. Therefore, option D is the most likely scenario to exacerbate carnivore-human conflicts.

Why the other options are less likely to exacerbate the conflicts:

Option A: Preventing wild animals from being attracted to human food sources would actually reduce carnivore-human conflicts by keeping animals from becoming habituated to humans. This would prevent potential issues, not exacerbate them.

Option B: The passage suggests that climate change could increase the frequency of human-carnivore encounters due to scarcity of resources. Therefore, addressing climate change would likely help prevent the issue rather than exacerbate it.

Option C: According to the passage, photographing wild animals in secured viewing areas is not a major driver of carnivore-human conflicts. Conflicts typically arise from behaviours that encourage animals to approach humans or interact in risky ways, not from observation in protected zones. Therefore, this is unlikely to exacerbate conflicts.

[▶ VIDEO SOLUTION](#)

LRDI

Explanation [25 - 29]:

We are given that the numbers keep increasing from left to right (clue 1), and the number keeps decreasing from top to bottom (clue 2)

The key takeaway from this is that 10 must be placed in Row 1, column 4, as placing it anywhere else would mean that the number above it, or right to it, must be greater than 10, which is not an option.

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1				10
Row 2				
Row 3				
Row 4				

Clue 3 says that one is either in the same row or the same column as 10.

The same logic that we used for 10 applies for 1; it must be either in Row 1, column 1, or Row 4, column 4; we don't know which one yet.

We are given that 2 and 3 are not in the same column or row as 10, meaning that they must occupy two spots from (Row 2, Col. 2), (Row 2, Col. 3), and (Row 3, Col. 3)

2 and 3 must be present in Row 2, column 2 or Row 3, column 3, as there is no number smaller than it to be in the cell left to it.

Clue 6 says that 4 and 6 are in the same row; this can be rows 1, 2, or 3.

Clue 5 is a good starting point.

Once we found the position of 10, the only positions possible for 9 are R1C3 or R2C4

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1			9	10
Row 2		2/3		
Row 3			2/3	
Row 4				

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1				10
Row 2		2/3		9
Row 3			2/3	
Row 4				

If 9 is placed in R2C4, 7 and 8 must be placed in row 1.

Clue 6 says that 4 and 6 are in the same row, but with 7 and 8 in row 1, no rows are left with two spaces. Hence, 9 can not be in R2C4, and the arrangement must be:

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1			9	10
Row 2		2/3		
Row 3			2/3	
Row 4				

7 and 8 must be in column 4, occupying R3C4 and R2C4, respectively.

Hence, 4, 6 must be in row 1:

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1	4	6	9	10
Row 2		2/3	5	8
Row 3			2/3	7
Row 4				1

25.4

The lowest sum is of the 4th row.

Therefore, is the correct answer.

[VIDEO SOLUTION](#)



26.A

We can see that the positions 10 and 1 are fixed and agree with the given statements.

Both the statements are true.

Therefore, Option A is the correct answer.

[VIDEO SOLUTION](#)

27.C

Positions 2 and 3 can be switched.

Either of them can be in column 2 or 3

Neither of the statements is necessarily true.

Therefore, Option C is the correct answer.

[VIDEO SOLUTION](#)

28.2

Placements of all the numbers except 2 and 3 can be determined.

Therefore, 2 is the correct answer.

[VIDEO SOLUTION](#)

29.26

The sum of numbers in the 4th column is $10+8+7+1 = 26$

Therefore, 26 is the correct answer.

[VIDEO SOLUTION](#)

30.B

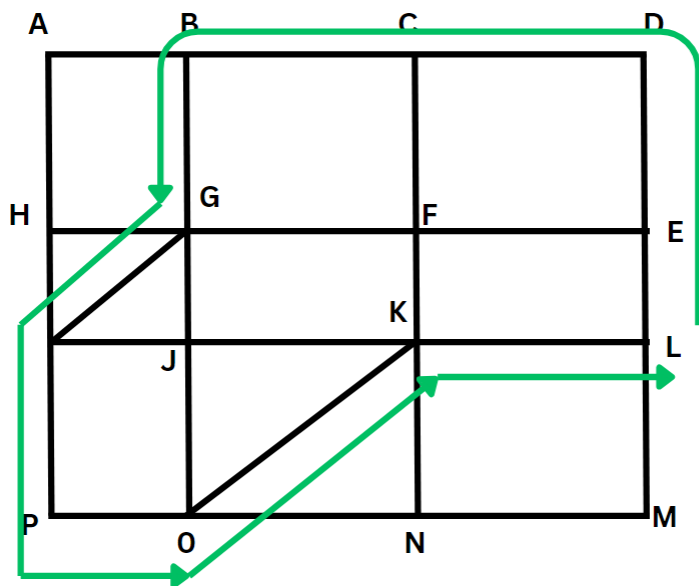
The first thing to realise here is the lengths of the paths.

KN should be equal to LM, giving the length of KO using Pythagoras theorem as 500m

Similarly, the length of HJ=OP=150m and length of GJ=EL=200m, giving the length of HG as 250 m

The shortest path from L to B and then to P (or the other way around would involve) using these hypotenuses as much as possible instead of the two adjacent sides. The shortest can be visualised as shown below or multitude of others variations, as there are multiple ways that would make one travel the shortest distance)

The below figure is the simplest one for visualisation.



Other possible paths are L-E-F-C... and following the same path.

The shortest distance in each of these instance would be $LE+ED+DC+CB+BG+GI+IP+PO+OK+KL$

Which would be $200+400+300+300+400+250+400+150+500+300 = 3200$

Therefore, Option B is the correct answer.

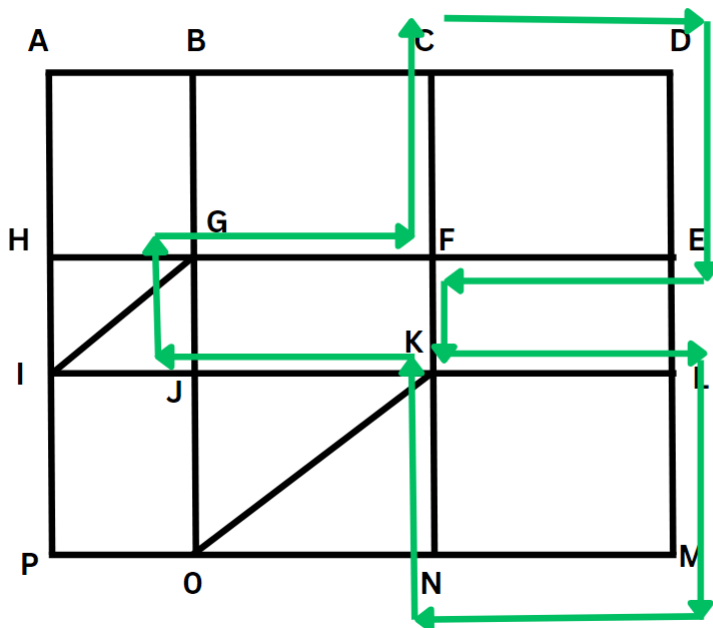
[VIDEO SOLUTION](#)



31. C

Since we can only walk along the side of lakes, that drastically reduces the paths we can take.

The diagram below shows the path with the maximum distance travelled.



The path is CD-DE-EF-FK-KL-LM-MN-NK-KJ-JG-GF-FC

(the reverse of this path is also valid)

We can either manually add the lengths or use shorter methods to note that in the path we travel walkways of length 400 m 4 times (DE, LM, KN, EF), walkways of length 300m 6 times (CD, EF, KL, MN, KJ, GF) and walkways of length 200 m 2 times (FK, JG)

Giving the total length to be $1600+1800+400 = 3800$

Therefore, Option C is the correct answer.

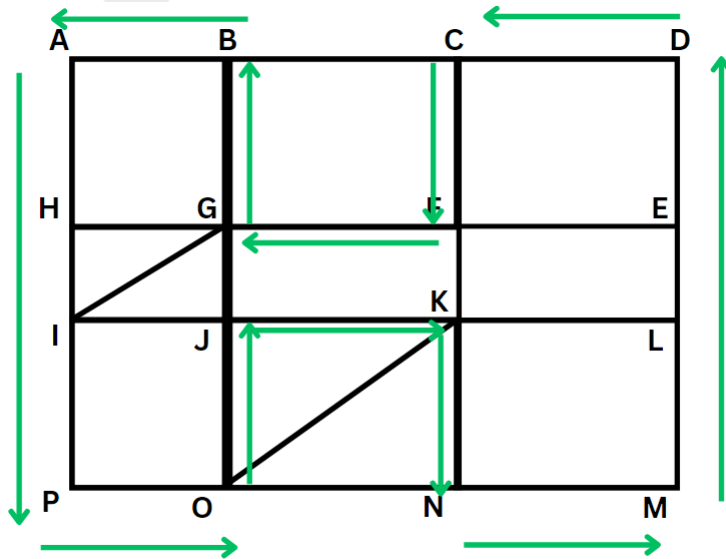
[VIDEO SOLUTION](#)

32. 5100

Counter to the first question, we should minimize the use of those hypotenuse walkways as they reduce the distance we travel.

The longest possible route would include then travelling through the edges of the rectangles and squares, and not touch any point more than once. Best case scenario would be us touching each point exactly once.

This can be visualized as:



Where we do not use any of the hypotenuse and cross through each gate exactly once.

The total distance would be:

$$AH+HI+ IP = 400+200+400 = 1000$$

$$DE+EL+LM = 400+200+400 = 1000$$

$$PO+JK+NM = 150+300+300 = 750$$

$$DC+FG+BA = 150+300+300 = 750$$

$$JO+KN = 400+400 = 800$$

$$CF+GB = 400+400 = 800$$

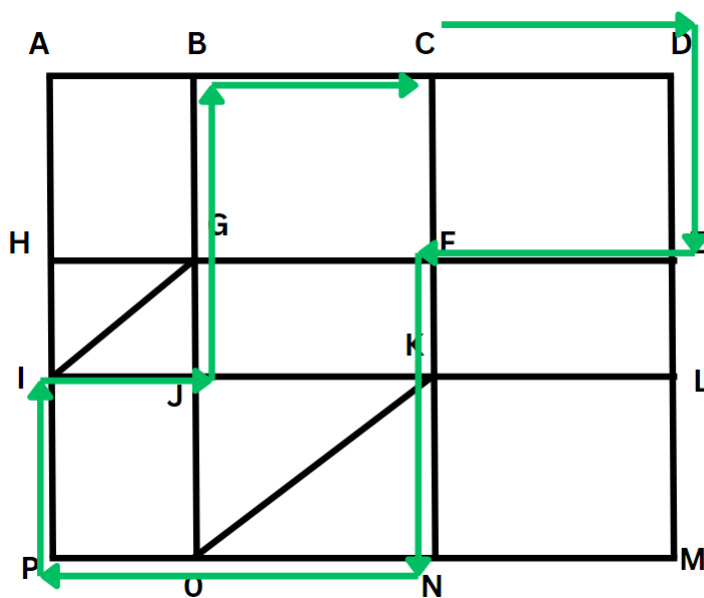
Adding up to: 5100m

Therefore, 5100 is the correct answer.

[VIDEO SOLUTION](#)

33. 3500

Similar to the previous question, we should try to avoid the hypotenuse



$$\text{The total path distance would be } 300(CD) + 400(DE) + 300(EF) + 200(FK) + 400(KN) + 300(NO) + 150(OP) + 400(PI) + 150(IJ) + 200(JG) + 400(GB) + 300(BC)$$

Adding up to 3,500 meters

Therefore, 3500 is the correct answer.

VIDEO SOLUTION

34.150

From the given chart, we can find the average rating on day 2:

$$\frac{(5 \times 1) + (10 \times 2) + (5 \times 3) + (20 \times 4) + (10 \times 5)}{5 + 10 + 5 + 20 + 10} = \frac{170}{50} = 3.4$$

We are given the cumulative average of day 1 and day 2 as 3.1, and the average at the end of day 1 is 3.

Let's take the number of ratings received on day 1 as x ; using this overall average and the average on day 2, we get the equation:

$$\frac{3x + (50 \times 3.4)}{x + 50} = 3.1$$

$$3x + 170 = 3.1x + 155$$

$$15 = 0.1x$$

$$x = 150$$

Therefore, the number of ratings received on day 1 is 150.

VIDEO SOLUTION

35.A

We are given that on day 3, a total of 100 ratings came in

The modes were 4 and 5, meaning that an equal number of 4 and 5 ratings came in; let it be $10b$ (since we are given that all ratings were non-zero multiples of 10)

Let's take the number of 1 and 2 ratings as $10a$ each, giving the number of 3 ratings as $20a$

Adding all of these up: $10a + 10a + 20a + 10b + 10b = 100$

$$40a + 20b = 100$$

$$2a + b = 5$$

The only integer combination for a and b , without them being zero, is a being 1 and b being 3 or $a=2$ and $b=1$

But taking b as 1 would make 3 as the mode rating, so we can not consider the latter case.

We are giving 10 - 1 ratings, 10 - 2 ratings, 20 - 3 ratings, 30 - 4 ratings, and 30 - 5 ratings.

$$\text{The average would be } \frac{(10 \times 1) + (10 \times 2) + (20 \times 3) + (30 \times 4) + (30 \times 5)}{100} = \frac{360}{100} = 3.6$$

Therefore, Option A is the correct answer.

VIDEO SOLUTION

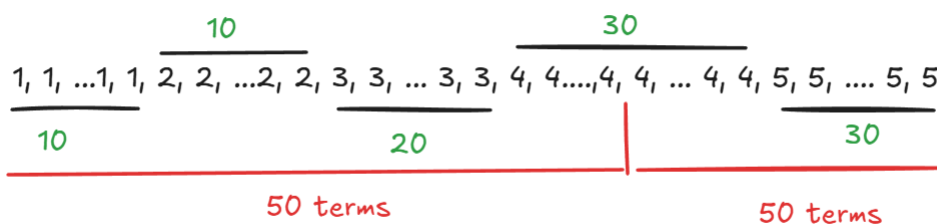


36.4

As deduced in the previous question, there were 10 1 ratings, 10 2 ratings, 20 3 ratings, 30 4 ratings, and 30 5 ratings.

The median would be the 50th and 51st ratings' average when arranged in ascending/descending order.

Both of which would be 4



Therefore, 4 is the correct answer

VIDEO SOLUTION

37.C

The cumulative average rating at the end of day 3 would be $\frac{(3.1 \times 200) + (3.6 \times 100)}{200 + 100}$

[the cumulative average rating on day 2 = 3.1

Number of ratings received by day 2 = 200]

$$\frac{360 + 620}{300} = \frac{980}{300} = 3.266$$

The increase in the cumulative average from day 2 to day 3 can be calculated as $\frac{3.266 - 3.1}{3.1} \times 100 \approx 5.34$

Which aligns with the statement given in option C.

Therefore, Option C is the correct answer.

VIDEO SOLUTION

38.C

Since we are given that the annual growth rate should be taken as constant, we can simply compare the ratio changes in the firm's PAT in 2019 and 2023 and compare them.

Firm A: 3000 to 3900; an increase of 900, giving the growth rate to be $\frac{900}{3000} = \frac{3}{10} = 0.3$

Firm C: 2400 to 3000; an increase of 600, giving the growth rate to be $\frac{600}{2400} = \frac{1}{4} = 0.25$

Firm E: 2400 to 3500; an increase of 1100, giving the growth rate to be $\frac{1100}{2400} = \frac{11}{24} \approx 0.45$

Firm B: 2800 to 3800; an increase of 1000, giving the growth to be $\frac{1000}{2800} = \frac{5}{14} \approx 0.35$

We can see that of the given ratios, the growth rate is highest for firm E (closest to 0.5). Therefore, the highest ARG, among the given options, will be for firm E. Therefore, Option C is the correct answer.

VIDEO SOLUTION

39. D

This can be a little difficult to solve since the bubbles are not closed within a fixed boundary, but in such instances, we should approximate the most visible cases.

Like the bubble of C in 2019 is almost 3 units vertically long in diameter, while that in 2023 is 2 units long.

The ratio of the amount spend on R&D on 2019 to 2024 would be given by: $\frac{2400\pi(1.5)^2}{3000\pi(1)^2} = \frac{8}{10} \times \frac{9}{4} = 9 : 5$

Which is option D.

VIDEO SOLUTION

40. D

The Pat per employee for the firm in 2023 will be:

Firm E: $\frac{3500}{1400} = \frac{35}{14} = \frac{5}{2}$

Firm A: $\frac{3900}{1300} = 3$

Firm F: $\frac{3200}{1000} = 3.2$

Firm C: $\frac{3000}{800} = \frac{15}{4} = 3.75$

The value if maximum for firm C, therefore, Option D is the correct answer.

VIDEO SOLUTION



41. B

We calculated the PAT per employee for F, C and E in the last question, using that information and calculating the PAT per employee for firm D to be $\frac{2400}{800} = 3$

For firms F, C and D, we can see that the area is the same hence the only factor to consider is PAT per employee

F is 3.2, C is 3.75, and D is 3

This leaves D as the minimum.

Comparing E and D (taking the proportionality constant as k)

E has PAT per employee as 2.5 with a radius of 1.5 units, the PRD value per employee would be $\frac{5}{2}k\pi \left(\frac{3}{2}\right)^2 = k\pi \times \frac{5 \times 9}{8}$

D has PAT per employee as 3, with a radius of 1 unit. the PRD value per employee would be $3k\pi (1)^2 = 3k\pi$

We can see that the value would be greater for E, therefore firm D will have the lowest R&D spending per employee in 2023

Therefore, Option B is the correct answer.

VIDEO SOLUTION

Explanation [42 - 46]:

We are told that each coach had at least two players. Clue 1 says that Xena trained more people than Yuki.

It must be the case that Yuki trained only two people. If Yuki had trained three, then Xena would have trained at least four players, leaving only one for Zara.

Hence, Yuki trained two people.

Coming to the scores given to the players themselves:

We are given that only 5 and 7 received a sam rating; everyone else received a distinct rating.

We are also given their average to be 4 (clue 4), giving the sum of all the scores they got to be 32

The sum of all numbers from 1 to 7 would be $\frac{8}{2} \times 7 = 28$, hence the repeated score must be $32 - 28 = 4$

Thus, the score of 5 and 7 was 4

Clue 5 says that player 2 got the highest score, which is 7

Clue 7 says that player 4 got a score double that of player 8 but less than player 5; the only possibility is player 4 getting 2 and player 8 getting 1.

Player	Score
1	
2	7
3	
4	2
5	4
6	
7	4
8	1

Now, considering clues 2, 3 and 6:

We are given the same coach trained in 5 and 7. And 2, 3, and 5 were trained by different coaches, with 1 and 4 being trained by the same coach. Cheer 6 informs us about the average number of players the coaches train.

We know that Yuki trained only two players.

Let's take the average score of Yuki's players to be x

The average of Xena's players would be $x/2$, and that of Zara's players would be $x-2$

The total score of Yuki's players would be $2x$.

The total score of Xena's players would be $3x/2$ or $2x$.

The total score of Zara's players would be $3x-6$ or $2x-4$

We have two cases to consider:



Let's say Xena and Zara had three players each.

The total score of Yuki's players would be $2x$.

The total score of Xena's players would be $3x/2$.

The total score of Zara's players would be $3x-6$.

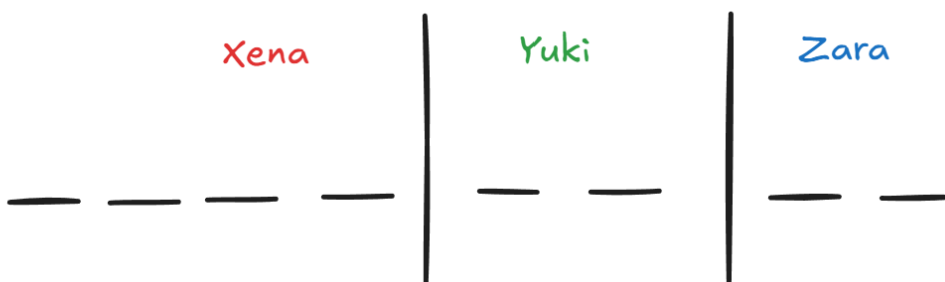
The sum of all these scores would be $\frac{13x}{2} - 6$ which should be equal to 32

This would give us the value of x as $\frac{13x}{2} = 38$

Which would give a non-integral value of $2x$, that is, the sum of Yuki's player's score.

Hence, this must not be the case.

The other possibility:



The total score of Yuki's players would be $2x$.

The total score of Xena's players would be $2x$.

The total score of Zara's players would be $2x-4$.

The sum of all these scores would be $6x-4$, which should be equal to 32

This would give the value of x as 6

Hence, the sum of all of Yuki's players would be 12

The sum of all of Xena's players would be 12

The sum of all of Zara's players would be 8

Yuki has only two players whose scores add up to 12; the only combination possible is scores 7+5, where seven were scored by player 2. Hence, player two must be under Yuki.

Zara got a total of 8 scores, with 7 and 5 gone. The combinations that could get this score are 2+6 and 4+4

The score of 2 is obtained by player 4, which must come with player 1

It is possible that 1 could have gotten a score 6

But then we run into a contradiction: players 3 and 5 would end up under the same coach.

Hence, Zara must have gotten 8 through $4+4$ with players 5 and 7.

All the remaining scores must be with Xena, adding up to 12, which is the case.

4 and 1 must be present together, and 3 must be present in Xena as well, giving us the arrangement.

	Xena				Yuki		Zara	
Players	$\frac{8}{1}$	$\frac{4}{2}$	$\frac{1/3}{3}$	$\frac{1/3}{6}$	$\frac{2}{7}$	$\frac{6}{5}$	$\frac{5}{4}$	$\frac{7}{4}$
Score:	1	2	3	6	7	5	4	4

42. B

We can see that Zara had two players.

Therefore, Option B is the correct answer.

[VIDEO SOLUTION](#)

43. 4

Score of player 7 was 4

Therefore, 4 is the correct answer.

[VIDEO SOLUTION](#)

44. 5

The rating of player 6 was 5

Therefore, 5 is the correct answer.

[VIDEO SOLUTION](#)

45. 6

We can determine the ratings of all the players except 1 and 3

Therefore, 6 is the correct answer.

[VIDEO SOLUTION](#)



46. B

Xena coached players numbered 1, 3, 4 and 8

Therefore, Option B is the correct answer.

[VIDEO SOLUTION](#)

Quant

47. 2160

Let the cost price of the item be C

We are given that Bina sells this at 19% loss or at $(1 - 0.19)C = 0.81C$ at 4860

This gives us the value of C at Rs. 6000

If Bina had sold this at 17% profit, the selling price would have been $1.17 \times 6000 = 7020$

So Shyam bought the product at 4860 and sold it to Hari at 7020

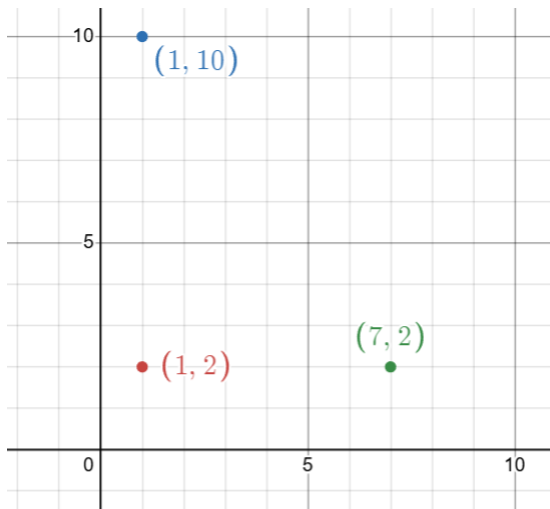
Giving the profit made by Shyam to be $7020 - 4860 = 2160$

Therefore, 2160 is the correct answer.

[VIDEO SOLUTION](#)

48. 2

Upon drawing a rough sketch of the coordinates given, we realise that this is a right-angled triangle.



The three side lengths are 6, 8 and 10 units

The inradius of a circle can be calculated using the formula: $rs = \text{Area}$, where s is the semi-perimeter of the triangle

The Area would be $\frac{1}{2} \times 6 \times 8 = 24$

and the semi-perimeter would be $\frac{10+6+8}{2} = 12$

Giving the inradius to be $\frac{24}{12} = 2$ units

Therefore, 2 is the correct answer.

[VIDEO SOLUTION](#)

49.340

(Note: This was a repeated question that appeared in both CAT 2023 and CAT 2024)

Let us assume the initial stock of all the fruits is S .

Let us take we have 'b' and 'a' mangoes initially.

Stock of Mangoes = 40% of $S = 2S/5$

The total number of fruits sold are Mangoes Sold + Apples Sold + Bananas Sold

$= 2S/10 + 96 + 4a/10 = S/2$ (Given)

$\Rightarrow S/5 + 96 + 2a/5 = S/2$

$\Rightarrow S = \frac{(4a + 960)}{3}$

$\Rightarrow \frac{4a}{3} + 320$

'a' has to be a multiple of 3 for the above term to be an integer.

But 'a' has to be a multiple of 5 for $4a/10$ to be an integer.

$\Rightarrow$ The smallest value of 'a' satisfying both conditions is 15.

$\Rightarrow \frac{4a}{3} + 320 = \frac{4(15)}{3} + 320 = 340$

Therefore, 340 is the correct answer.

[VIDEO SOLUTION](#)

50.14

The first term of the expression can be rewritten as $\frac{\frac{1}{3} \log_2(a+b)}{\log_2 c}$

Using the property $\frac{m}{n} \log_a b = \log_a b^{\frac{m}{n}}$ this can be rewritten as

$$\frac{\log_2(a+b)^{\frac{1}{3}}}{\log_2 c}$$

And finally using the property $\frac{\log_b a}{\log_b c} = \log_c a$, we can rewrite the expression as

$$\log_c(a+b)^{\frac{1}{3}}$$

Doing identical operations in the second term, we get the entire left-hand side to be:

$$\log_c(a+b)^{\frac{1}{3}} + \log_c(a-b)^{\frac{1}{3}}$$

Using property $\log_c a + \log_c b = \log_c(ab)$ we get

$$\log_c \left[(a+b)^{\frac{1}{3}} (a-b)^{\frac{1}{3}} \right]$$

$$\log_c \left[(a+b)(a-b) \right]^{\frac{1}{3}}$$

$$\log_c \left[(a^2 - b^2) \right]^{\frac{1}{3}}$$

This expression is given to be equal to $2/3$

Using the definition of log: $\log_c N = a$ which is $c^a = N$

we get: $c^{\frac{2}{3}} = (a^2 - b^2)^{\frac{1}{3}}$

Cubing both sides:

$$c^2 = a^2 - b^2$$

Finally giving $a^2 = b^2 + c^2$

We have upper limits on b and c as 10, and we want to maximize the value of a squared.

This can be thought of as a right-angled triangle, and the value of a will be maximum when both b and c are equal to 10, giving $a^2 = 200$, but this would not give an integer value of a

We need to adjust a^2 to the biggest square less than 200, which is 196

Giving the value of a as 14.

Therefore, 14 is the correct answer.

[VIDEO SOLUTION](#)

100+ CAT Quant Questions



51. A

Looking at the additional information about the prime numbers should make one realise that they are the key to solving the question.

$f(16000)$ can be written as $f(2^8 \times 5^4)$

Now, we can try to find these individual values:

For any prime p: $f(p)=1$

$$f(p^2) = f(p) f(p) + f(p) + f(p) = 1 + 1 + 1 = 3$$

$$f(p^3) = f(p^2) f(p) + f(p^2) + f(p) = 3 + 3 + 1 = 7$$

This way, we can find the function output for any prime number raised to a power.

We can see that each new exponent is twice the previous output +1, solving this way till prime raised to power 8

$$f(p^4) = 7 + 7 + 1 = 15$$

$$f(p^5) = 15 + 15 + 1 = 31$$

$$f(p^6) = 31 + 31 + 1 = 63$$

$$f(p^7) = 63 + 63 + 1 = 127$$

$$f(p^8) = 127 + 127 + 1 = 255$$

Using these values in the original expression of $f(2^8 \times 5^4) = f(2^8) f(5^4) + f(2^8) + f(5^4)$ we get

$$f(2^8 \times 5^4) = (255 \times 15) + 255 + 15 = 4095$$

Therefore, Option A is the correct answer.

[VIDEO SOLUTION](#)

52. B

From the sum and product of roots, we get: $\alpha + \beta = -\frac{\lambda}{3}$ and $\alpha\beta = -\frac{1}{3}$

Simplifying the expression given in the question, we get: $\frac{\alpha^2 + \beta^2}{\alpha^2\beta^2} = 15$

and substituting the denominator's value as $1/9$, we get: $\alpha^2 + \beta^2 = \frac{15}{9}$

We want the expression $\alpha^3 + \beta^3$, so multiplying both sides by $\alpha + \beta$, we get:

$$\alpha^3 + \beta^3 + \alpha\beta(\alpha + \beta) = \frac{15}{9}(\alpha + \beta)$$

$$\alpha^3 + \beta^3 + \frac{\lambda}{9} = \frac{15}{9}\left(-\frac{\lambda}{3}\right)$$

$$\alpha^3 + \beta^3 + \frac{\lambda}{9} = -\frac{5\lambda}{9} - \frac{\lambda}{9} = -\frac{2\lambda}{3}$$

We would still need to find the value of λ

This we can do from the initial relation we had:

$$\alpha^2 + \beta^2 = \frac{15}{9}$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \frac{15}{9}$$

$$\frac{\lambda^2}{9} + \frac{2}{3} = \frac{15}{9}$$

$$\frac{\lambda^2}{9} = \frac{15-6}{9} = \frac{9}{9} = 1$$

This would finally give us $\lambda^2 = 9$

Using this in our required expression, we get:

$$(\alpha^3 + \beta^3)^2 = \left(-\frac{2\lambda}{3}\right)^2 = \frac{4 \times 9}{9} = 4$$

Therefore, Option B is the correct answer.

[VIDEO SOLUTION](#)

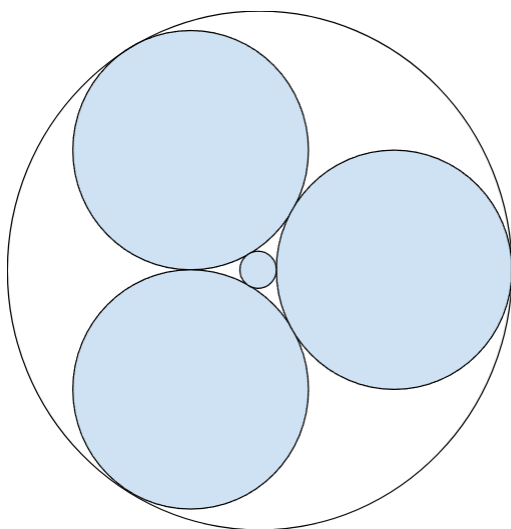
53. C

Let's take Rajesh and Garima's ages to be R and G , respectively
 From the given ratio, we can see that Rajesh is older than Garima, so let's take $R=G+x$
 When Rajesh was of age G , which was x years ago, Garima was of $G-x$ years old
 Giving the ratio as $\frac{G}{G-x} = \frac{3}{2}$
 This gives us G as $3x$, which in turn gives R as $4x$
 We are asked the ratio when Gramia becomes $4x$ years old.
 By that time, Rajesh will be $5x$ years old.
 Giinv their ratio as $\frac{5x}{4x} = 5 : 4$
 Therefore, Option C is the correct answer.

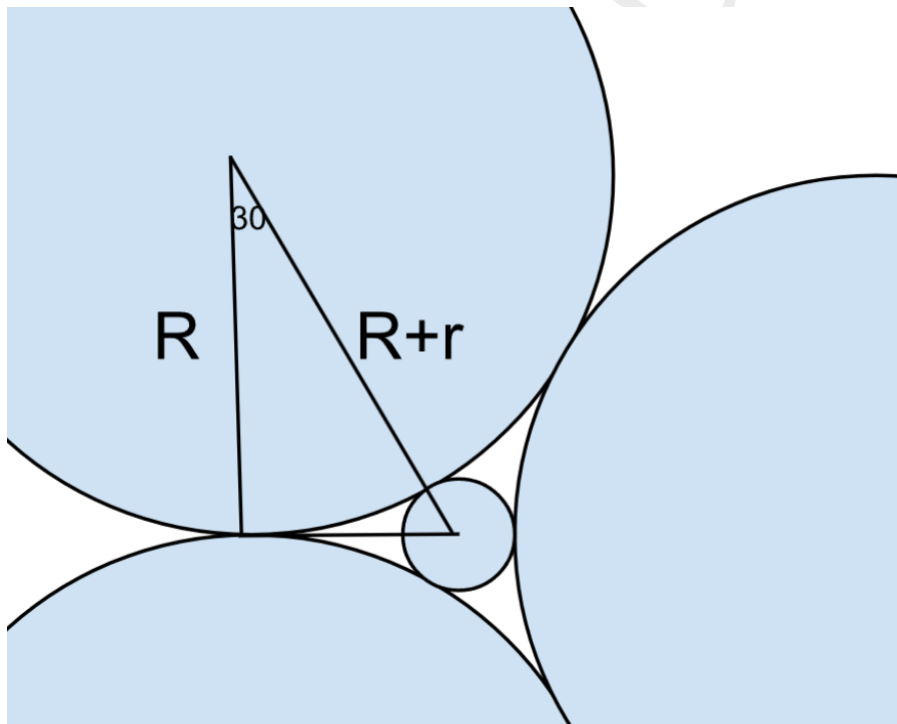
VIDEO SOLUTION

54. A

Let's take the radius of the original circles to be R and that of the circle in between the three circles to be r .



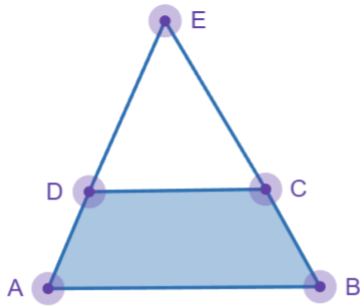
Joining the centres of the three circles, we will get an equilateral triangle of length $2R$.
 The distance between the circle's centre and the original circle's centre would be $R+r$.



Using this right angle triangle, we can get the relation: $\frac{R}{R+r} = \frac{\sqrt{3}}{2}$
 We can take $R = \sqrt{3}a$ and $R+r$ as $2a$, this would give us r as $R = (2 - \sqrt{3})a$
 The outer circle will have a radius of $2R+r$
 We need to find the ratio of $\frac{2R+r}{r}$
 This will be equal to $\frac{(2\sqrt{3}+2-\sqrt{3})a}{(2-\sqrt{3})a} = \frac{2+\sqrt{3}}{2-\sqrt{3}} = 4 + 3 + 4\sqrt{3} = 7 + 4\sqrt{3} : 1$
 Therefore, Option A is the correct answer.

55. A

The simplest way to visualise this would be with symmetry.



We can take AD to be x and CB to be $3-x$; since the perimeter of ABCD is 6, and AB and CD are given as 2 and 1 cm, respectively, that leaves $AD+BC$ as three only.

We can see that triangles AEB and DEC are similar, with the lengths of AB being double of CD, essentially making D the mid-point of AE and C the mid-point of EB.

Through this, we get the length of DE and CE to be x and $3-x$.
Give us the AE and BE lengths as $2x$ and $6-2x$, respectively.

Giving the perimeter of AEB as $2x+6-2x+2 = 8$ cm

Therefore, Option A is the correct answer.

100+ CAT DILR Questions



56. A

We can divide the list into 4 elements: the first 10 as a , the next 10 as b , the next 10 as c , and the last 10 as d

From the relations we are given, we can form the equations: $\frac{a+b+c}{3} = 40,000$
 $\frac{b+c+d}{3} = 60,000$ and $\frac{a+d}{2} = 50,000$

Adding the first two equations, we get $a + 2(b + c) + d = 300,000$

We can substitute the value of $a+d$ as 100,000 to get $b+c$ as 100,000

Using this value in the first and second equation would give a and d as 20,000 and 80,000, respectively.

We are told that the average of the first 10 employees increases by 100%, that is, it changes from 20,000 to 40,000

The average of the last 10 increases by 200%; that is, it changes from 80,000 to 240,000

The total of all the four elements would be $40,000+100,000+240,000 = 380,000$

Giving the average to be $\frac{380,000}{4} = 95,000$

Therefore, Option A is the correct answer.

57. 139

We can take the work done by Amal, Vimal and Sunil to be A , V and S , respectively.

Let's take the total work they did to be T .

We are given the equations:

$$150A + 150V = T \quad \dots(1)$$

$$100V + 100S = T \quad \dots(2)$$

$$75A + 135V + 45S = T \quad \dots(3)$$

$$\text{Adding (1) and (2), we get: } 150A + 250V + 100S = 2T \quad \dots(4)$$

$$\text{And multiplying (3) with 2 we get: } 150A + 270V + 100S = 2T \quad \dots(5)$$

$$\text{Subtracting (5) from (4), we get } 10S = 20V \text{ or simply } S = 2V$$

Using this in (2), we get the total work T to be $300V$, and using that result in (1), we get $A=V$

Therefore, the work done by A , V and S equals V , V , and $2V$ units per day.

Now, in the question, we are given work cycles. To simplify the calculations, we should consider a time duration that is the LCM of the period taken by the three agents, which in this case would be 6

In 6 days, Amal will work for 6 days, doing $6V$ units of work.

Vimal will work for 3 days, doing $3V$ units of work.

Sunil will work for 2 days, doing $4V$ units of work.

So, in one 6-day cycle, 13V units of work will be done.
 Dividing the total work (300V) by 13V we can see that in 23 cycles 299V units of work will be done.
 These 23 cycles will be $23 \times 6 = 138$ days
 The remaining 1V units of work will be done the next day.
 Therefore, a total of 139 days will be required.

[VIDEO SOLUTION](#)

58. A

There are two critical points for the inequality to consider: $x = -5$ and $x = 3/2$



Region I: x is greater than $3/2$

In this scenario, both the terms would be positive; cross-multiplying, we get the relation $2x - 3 \leq x + 5$

Giving the boundary $x \leq 8$, hence giving us the valid range as $\frac{3}{2} < x \leq 8$

Region II: $-5 < x < \frac{3}{2}$

In this case, the right-hand side will be a negative value, and hence, the sign would change when multiplying, giving the inequality

$$2x - 3 \geq x + 5$$

Which will give $x > 8$, which is out of bounds for this region

Another way is to put a value in the region to check for the validity of the inequality; by putting $x = 0$, we could see that the inequality does not hold in this region

Region III: x less than -5

In this scenario, both the terms are negative, essentially giving us the same boundary as region 1; we take the lower bounds, giving us that x has to be less than -5

Therefore, for the given inequality to hold true $x < -5$ or $\frac{3}{2} < x \leq 8$

Hence, Option A is the correct answer.

[VIDEO SOLUTION](#)

59. D

Let's take the amount invested by Sunil to be X .

The amount received by Anil at the end of 6 years would be $22000 \left(1 + \frac{4}{2 \times 100}\right)^{6 \times 2} = 22000 (1.02)^{12}$

The amount received by Sunil at the end of 5 years would be $X (1.02)^{10}$

In the 6th year, Sunil invests this at a simple interest of 10%, giving him an interest of $X (1.02)^{10} \times 0.1$

Giving the total amount with him at the end of 6 years to be $X (1.02)^{10} \times (1 + 0.1)$

Equating the final amount with Sunil and Anil, we get:

$$X (1.02)^{10} \times (1.1) = 22000 (1.02)^{12}$$

$$X = \frac{22000(1.02)^2}{1.1} = 20808$$

Therefore, Option D is the correct answer.

[VIDEO SOLUTION](#)

60. A

Let's take the scheduled time taken by the bus to be t

From the first statement (bus travelling at 60 kmph), we can get the total distance travelled by bus to $60(t+3.5)$

The second scenario gives us that the bus covered two-thirds of the distance in one-third of the time, meaning that the remaining one-third distance was covered in two-thirds of the time, giving us the relation $\frac{1}{3}st$ covered in $\frac{2}{3}t$ giving the speed to be $\frac{s}{2}$ which is given as 40 km/h, thereby giving the usual speed of the bus to be 80 km/hr

Now the first relation we get $60(t+3.5)=80t$

Giving us $t=10.5$ hours

Thus, the bus usually takes 10.5 hours on its journey.

Starting at 9:00, it will complete the journey at 7:30 pm

Therefore, Option A is the correct answer.

[VIDEO SOLUTION](#)

100+ CAT VARC Questions



61. D

We must bring the right-hand side in the form so that everything has the same power.

25 has factors 1, 5 and 25

The only common factor 40 and 25 have is 5 (other than 1 of course, which does not work)

So the right-hand side can be rewritten as $(2^5)^5 \times (3^8)^5$

$$(32 \times 81 \times 81)^5$$

$$(209952)^5$$

Giving the value of $m - n$ as $209952 - 5 = 209947$

Therefore, Option D is the correct answer.

VIDEO SOLUTION

62. A

There are multiple ways of solving these sorts of questions. One method is to look for powers of the term in the numerator that leave a remainder of 1 or -1 when divided by the denominator.

Noting down the powers of 3, 3, 9, 27, 81, 243

243 is one such number, 242 is multiple of 11 (11 times 22), hence 243 will leave a remainder of 1 when divided by 11.

243 is 3 raised to power 5; we can rewrite the given term as $\frac{3^{330} \times 3^3}{11}$

The overall remainder will be $\left[\frac{3^{330}}{11}\right]_R \times \left[\frac{3^3}{11}\right]_R$

$$\left[\frac{3^{5 \times 66}}{11}\right]_R \times \left[\frac{3^3}{11}\right]_R$$

$$\left[\frac{243^{66}}{11}\right]_R \times \left[\frac{3^3}{11}\right]_R$$

$$1^{66} \times \left[\frac{27}{11}\right]_R$$

$$1 \times 5$$

$$5$$

Therefore, Option A is the correct answer.

VIDEO SOLUTION

63. 7

In such questions, we should be trying to complete the squares.

We see a xy term; we need to accommodate that in a square that has both x and y terms.

Since there is only one other term with x , we also need to have it entirely in the square.

$$(2x - y)^2 = 4x^2 + y^2 - 4xy$$

Using this in the given equation, we are left with $(2x - y)^2 + 3y^2 + 3 - 6y$

This can be written as $(2x - y)^2 + 3(y^2 + 1 - 2y)$

$$(2x - y)^2 + 3(y - 1)^2 = 0$$

Since both the squares add up to 0, this is only possible when the squares themselves are 0

This would give us $y=1$ from the second term, and using that, we get $x=1/2$ from the first term.

Therefore the value of $4x+5y$ will be $2+5=7$

Hence, 7 is the correct answer.

VIDEO SOLUTION

64. 11

Squaring on both sides, we get:

$$x + 6\sqrt{2} + x - 6\sqrt{2} - 2(x^2 - 72)^{\frac{1}{2}} = 8$$

$$x - (x^2 - 72)^{\frac{1}{2}} = 4$$

Bringing x to the other side, we get:

$$-(x^2 - 72)^{\frac{1}{2}} = 4 - x$$

Squaring on both sides again, we get:

$$x^2 - 72 = 16 + x^2 - 8x$$

$$8x = 88$$

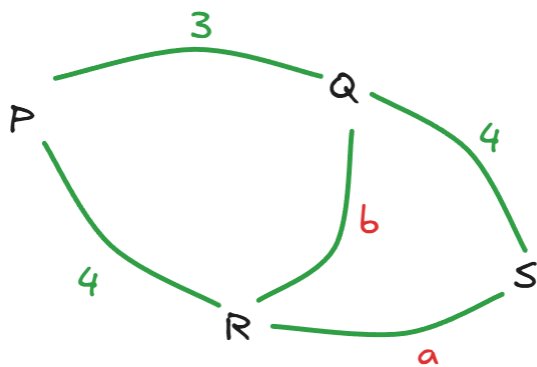
$$x = 11$$

Therefore, 11 is the correct answer.

VIDEO SOLUTION

65. 7

Let's take the number of paths between Q and R to be b and the number of paths between R and S to be a



We are given the paths from P to S through R (which would be $4a$), the paths from P to S through Q (which would be 12) and the paths from P to Q to R to S, which would be $3ab$) is equal to 62

Giving the relation $4a+12+3ab = 62$

Or $4a+3ab = 50$

The paths from Q to R directly (which would be b), through P (which would be 12) and through S (which would be $4a$) are 27

Giving the relation $b+12+4a = 27$

Or $4a+b = 15$

Subtracting this equation from the first one we got, we get $3ab-b=35$, or $b(3a-1)=35$

b can be $1, 3, 5$ or 7

Substituting these values in the second equation, we see that it can not be 1 or 5 , leaving only 3 or 7 as the possible values.

Substituting b as 3 in the first equation would give $13a=50$, which is not true.

Substituting b as 7 in the first equation would give $25a = 50$, which would give $a=2$

We are asked the number of paths from Q to R, which is $b=7$

Therefore, 7 is the correct answer.

[VIDEO SOLUTION](#)

CAT Expected Questions PDF



66. B

We can consider the quadrants of a graph:

First quadrant: Both x and y are positive

This would change the equation to $2x+y=15$ and $x=20$, giving a negative value of y ; hence, this is not the case.

Second quadrant: x is negative, but y is positive

This would change the equations to $y=15$ and $x=20$, giving a positive value of x , which hence can not be the case.

Third quadrant: Both x and y are negative

This would change the equation to $y=15$ and $x-2y=20$; this gives a positive value of y and hence can not be the case.

Fourth quadrant: x is positive, but y is negative

This would change the equations to $2x+y=15$ and $x-2y=20$; this gives the value of x as 10 and y as -5 , which would lie in the fourth quadrant.

The value of $x-y$ would be $10-(-5)=15$

Therefore, Option B is the correct answer.

[VIDEO SOLUTION](#)

67. B

Let's start from the step when there was 50% concentration.

Let's take there to be $2T$ solution: T acid and T water.

Adding 15 litres of acid increases the acid concentration to 80% , giving the equation $\frac{T+15}{2T+15} = \frac{4}{5}$

Solving this would give us $T=5$

This means that there were 5 litres of acid and 5 litres of water after mixing 2 litres of water.

Therefore, there would be 5 litres of acid and 3 litres of water before adding the water.

We are asked the ratio of water to acid, which would be $3:5$

Therefore, Option B is the correct answer.

[VIDEO SOLUTION](#)

68. B

Opening the brackets, we get the series as: $\left(\frac{1}{5}\right)^2 - \left(\frac{1}{5} \times \frac{1}{7}\right) + \left(\frac{1}{5}\right)^4 - \left(\frac{1}{5} \times \frac{1}{7}\right)^2 + \left(\frac{1}{5}\right)^6 - \left(\frac{1}{5} \times \frac{1}{7}\right)^6 + \dots$

These are two infinite GPs when rearranged:

$$\left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^4 + \left(\frac{1}{5}\right)^6 + \dots - \left(\frac{1}{5} \times \frac{1}{7}\right) - \left(\frac{1}{5} \times \frac{1}{7}\right)^6 - \left(\frac{1}{5} \times \frac{1}{7}\right)^2 - \dots$$

The sum of the first series would be $\frac{\frac{1}{25}}{1 - \frac{1}{25}} = \frac{1}{24}$

The sum of the second series would be $\frac{\frac{1}{35}}{1 - \frac{1}{35}} = \frac{1}{34}$

The answer to the given series would then be $\frac{1}{24} - \frac{1}{34} = \frac{10}{816} = \frac{5}{408}$

Therefore, Option B is the correct answer.

[VIDEO SOLUTION](#)